VISPA Crawling Locomotion Controller

Two-Stage Hierarchical Whole-Body Control
for a Multi-Arm Orbital Robot Servicer

Software Documentation & Architecture Reference

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Abstract

This document describes the software implementation of a two-stage hierarchical controller for whole-body crawling locomotion of the VISPA multi-arm orbital robot servicer. Stage 1 is a **Centroidal NMPC** (20 Hz) that plans center-of-mass trajectories and contact wrenches. Stage 2 is a **Whole-Body QP** (125 Hz) that tracks the NMPC references while satisfying the full robot dynamics, joint limits, and end-effector swing constraints. The controller is validated in a MuJoCo closed-loop simulation with HOTDOCK-style weld constraints.

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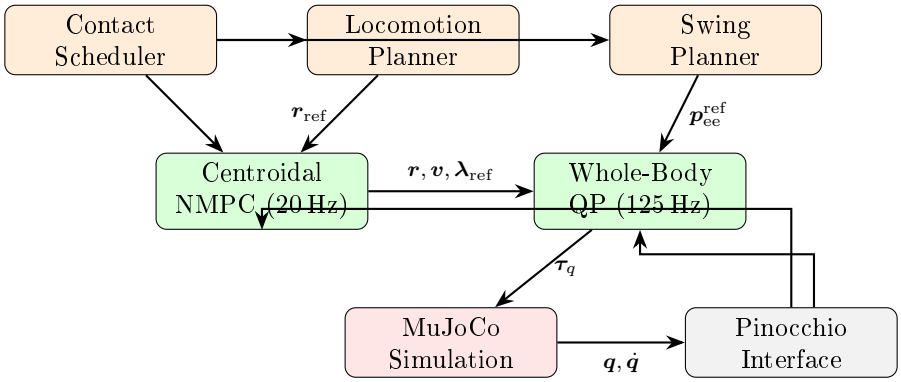
1 System Overview

1.1 Robot Description

VISPA is a dual-arm orbital servicing robot designed for crawling locomotion along a truss structure. Each arm has 6 revolute joints and a HOTDOCK gripper.

Property	**Value**
Generalized coordinates n_q	19 (quaternion + position + 12 joints)
Velocity dimension n_v	18 (6 torso twist + 12 joint velocities)
Actuators n_u	12 (6 per arm)
Total mass	31.68 kg
Gravity	0 (orbital microgravity)

1.2 Architecture



2 Mathematical Formulation

2.1 Stage 1: Centroidal NMPC

The NMPC plans the CoM trajectory over a receding horizon N=20, Δt=0.05 s (1 s horizon).

State & control. $\boldsymbol{x} = [\boldsymbol{r}_{\text{com}}^\top, \boldsymbol{v}_{\text{com}}^\top, \boldsymbol{L}_{\text{com}}^\top]^\top \in \mathbb{R}^9$, $\boldsymbol{u} = [\boldsymbol{f}_1^\top, \boldsymbol{\tau}_1^\top, \boldsymbol{f}_2^\top, \boldsymbol{\tau}_2^\top]^\top \in \mathbb{R}^{12}$.

Centroidal dynamics.

$$m\dot{\boldsymbol{v}} = \textstyle\sum_j \boldsymbol{f}_j, \qquad \dot{\boldsymbol{L}} = \textstyle\sum_j (\boldsymbol{p}_j - \boldsymbol{r}) \times \boldsymbol{f}_j + \boldsymbol{\tau}_j. \tag{1}$$

Momentum exchange. $\boldsymbol{h}_w^{k+1} = \boldsymbol{h}_w^k - \Delta \boldsymbol{L}_{\text{robot}}^k$, with $\boldsymbol{h}_w^{\min} \le \boldsymbol{h}_w^k \le \boldsymbol{h}_w^{\max}$.

Cost. $\ell(\boldsymbol{x}, \boldsymbol{u}) = \|\boldsymbol{r} - \boldsymbol{r}_{\text{ref}}\|_{\boldsymbol{W}_r}^2 + \|\boldsymbol{v} - \boldsymbol{v}_{\text{ref}}\|_{\boldsymbol{W}_v}^2 + w_f \sum_j \|\boldsymbol{f}_j\|^2 + w_\tau \sum_j \|\boldsymbol{\tau}_j\|^2$.
Discretized with RK4, solved by CasADi/IPOPT with warm-starting. Typical: 16–30 ms.

2.2 Stage 2: Whole-Body QP

Decision variables. $\boldsymbol{z} = [\ddot{\boldsymbol{q}}_t^\top, \ddot{\boldsymbol{q}}^\top, \boldsymbol{\lambda}^\top, \boldsymbol{\tau}_q^\top]^\top$, with $\ddot{\boldsymbol{q}}_t \in \mathbb{R}^6$ (torso), $\ddot{\boldsymbol{q}} \in \mathbb{R}^{12}$ (joints), $\boldsymbol{\lambda} \in \mathbb{R}^{12}$ (wrenches), $\boldsymbol{\tau}_q \in \mathbb{R}^{12}$ (torques).

Equality constraints.

1. Full dynamics: $\boldsymbol{H}\ddot{\boldsymbol{q}}_{\text{robot}} + \boldsymbol{C} = \boldsymbol{B}_u \boldsymbol{\tau}_q + \boldsymbol{J}_c^\top \boldsymbol{\lambda}$
2. Contact rigidity: $\boldsymbol{J}_c \ddot{\boldsymbol{q}}_{\text{robot}} = -\dot{\boldsymbol{J}}_c \dot{\boldsymbol{q}}_{\text{robot}}$

Task hierarchy (weighted single QP via qpOASES):

Prio.	Task	α	Equation
1	CoM tracking	10^3	$\boldsymbol{J}_{\text{com}}\ddot{\boldsymbol{q}} = \ddot{\boldsymbol{r}}^{\text{des}}_{\text{com}} - \dot{\boldsymbol{J}}_{\text{com}}\dot{\boldsymbol{q}}$
2	EE tracking	$5{\times}10^2$	$\boldsymbol{J}_{\text{ee}}\ddot{\boldsymbol{q}} = \ddot{\boldsymbol{p}}^{\text{des}}_{\text{ee}} - \dot{\boldsymbol{J}}_{\text{ee}}\dot{\boldsymbol{q}}$
3	Posture	10^2	$\ddot{\boldsymbol{q}} = K_p(\boldsymbol{q}_{\text{nom}} - \boldsymbol{q}) - K_d\dot{\boldsymbol{q}}$
4	Wrench	10^1	$\boldsymbol{\lambda} = \boldsymbol{\lambda}_{\text{ref}}$
5	Torque min.	10^0	$\boldsymbol{\tau}_q = \boldsymbol{0}$
6	Regularization	10^{-2}	$\ddot{\boldsymbol{q}}_{\text{robot}} = \boldsymbol{0}$

PD feedback for CoM and EE:

$$\ddot{\boldsymbol{r}}^{\text{des}}_{\text{com}} = \boldsymbol{a}_{\text{ff}} + \boldsymbol{K}^c_p(\boldsymbol{r}_{\text{ref}} - \boldsymbol{r}) + \boldsymbol{K}^c_d(\boldsymbol{v}_{\text{ref}} - \boldsymbol{v}), \tag{2}$$

$$\ddot{\boldsymbol{p}}^{\text{des}}_{\text{ee}} = \boldsymbol{K}^e_p(\boldsymbol{p}_{\text{ref}} - \boldsymbol{p}_{\text{ee}}) + \boldsymbol{K}^e_d(\dot{\boldsymbol{p}}_{\text{ref}} - \dot{\boldsymbol{p}}_{\text{ee}}). \tag{3}$$

2.3 Swing Arm Trajectory

During single-support, the free arm follows:

$$\boldsymbol{p}(\tau) = \boldsymbol{p}_{\text{start}} + \Delta\boldsymbol{p}\, s(\tau) + d_{\text{clear}}\, \hat{\boldsymbol{n}}\, \sin^2(\pi\tau), \qquad \tau \in [0, 1], \tag{4}$$

where $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ is a rest-to-rest quintic. Both s and $\sin^2(\pi\tau)$ have zero first and second derivatives at boundaries, ensuring smooth detach/attach.

3 Gait Scheduling

Alternating DS (0.5 s) and SS (1.0 s). For 4 steps from anchors (2,2):

Phase	Type	Time [s]	A@	B@	Swing
0	DS	$[0.0, 0.5)$	2	2	—
1	SS_A	$[0.5, 1.5)$	2	2→3	B: x=−0.4 → +0.4
2	DS	$[1.5, 2.0)$	2	3	—
3	SS_B	$[2.0, 3.0)$	2→3	3	A: x=−0.4 → +0.4
		$\cdots$ (continues for 6.5 s total)			

4 MuJoCo Integration

4.1 Frame Alignment

Anchor positions are read from MuJoCo sites via `read_anchors_from_mujoco(mj_model, mj_data)`, eliminating frame mismatch.

4.2 Weld Constraints

HOTDOCK grippers modeled as MuJoCo equality constraints. At runtime, `mj_data.eq_active[i]` is toggled per phase. Re-docking requires a proximity check (< 10 cm).

4.3 State Conversion

`pinocchio_to_mujoco()` and `mujoco_to_pinocchio()` handle: dual freejoints (MuJoCo n_q=26, Pinocchio n_q=19), quaternion convention ($[w, x, y, z]$ vs. $[x, y, z, w]$), structure body state.

5 Test Results

5.1 Unit Validation (Phases 1–4)

Phase	Test	Result
1. RobotInterface	17 dimension/rank checks	PASS
2. IK + Docking	Tool error $< 10^{-9}$ m	PASS
3. NMPC standalone	Warm: 16.5 ms, 5 IPOPT iter	PASS
4. QP standalone	Dynamics residual: 1.6×10^{-13}	PASS

5.2 Closed-Loop (Phase 5)

Regulation (1 s, DS). 20/20 NMPC + 120/120 QP. Error: 0.11 m. Contact violation: 51 μm.

Full locomotion (6.5 s). 122/130 NMPC + 780/780 QP. **Controller computes; trajectory diverges after first SS** (see Sec. 6).

Solver	Budget	Mean	Max	Success
NMPC	50 ms	29.6 ms	50.3 ms	93.8%
QP	8 ms	5.6 ms	31.5 ms	100%

6 Known Issues & Tuning Roadmap

1. **Weld re-activation failure.** Swing arm does not reach target ($\|\boldsymbol{e}_{\text{ee}}\| \approx 0.37$ m at mid-swing). Weld never re-activates; robot floats free.

   Tuning levers: increase $\tau_{\max}$ (currently 10 Nm); increase swing duration (1 s $\rightarrow$ 2–3 s); reduce clearance/distance; increase α_{ee} during SS.

2. **NMPC horizon clips transitions.** 1 s horizon sometimes spans the SS$\rightarrow$DS boundary with incorrect contact mode at tail.

3. **Simplified AOCS.** First-order wheel momentum integrator; proper CMG model needed.

4. **Structure drift.** Freejoint in MuJoCo (5 mm/s). Fix: increase structure mass or use `mocap` body.

7 File Reference

File	Description
`robot_interface.py`	Pinocchio wrapper: CRBA, RNEA, Jacobians, $\dot{J}\dot{q}$
`contact_scheduler.py`	Gait phases, `plan_traversal()`, MuJoCo anchor reader
`locomotion_planner.py`	CoM reference trajectory (quintic interpolation)
`swing_planner.py`	Swing arm Cartesian trajectory
`ik.py`	Inverse kinematics: `dock_configuration()`
`simulation_loop.py`	MuJoCo closed-loop, weld management, nested loops
`convert.py`	Pinocchio ↔ MuJoCo state conversion
`solvers/centroidal_nmpc.py`	CasADi/IPOPT NMPC with warm-start
`solvers/wholebody_qp.py`	qpOASES whole-body QP, 6-level hierarchy
`solvers/hierarchical_qp.py`	Generic weighted/strict hierarchical QP
`solvers/contact_phase.py`	Contact data structures
`solvers/nmpc_solver.py`	Generic CasADi NMPC wrapper
`test_integration.py`	Phase 1–4 validation (14 tests)

8 Dependencies

Library	Role	Version
Pinocchio	Rigid-body dynamics	3.x
CasADi	NMPC + IPOPT	3.6.x
qpOASES	Active-set QP	3.2
MuJoCo	Physics simulation	3.2.x
NumPy	Linear algebra	1.26+

9 Quick Start

```
from robot_interface import RobotInterface
from contact_scheduler import ContactScheduler, read_anchors_from_mujoco
from locomotion_planner import LocomotionPlanner
from swing_planner import SwingPlanner
from ik import dock_configuration
from solvers.centroidal_nmpc import CentroidalNMPC, CentroidalNMPCConfig
from solvers.wholebody_qp import WholeBodyQP, WholeBodyQPConfig
import mujoco, numpy as np

# 1. Load models
robot = RobotInterface("VISPA_crawling_fixed.urdf", gravity='zero')
mj_model = mujoco.MjModel.from_xml_path("VISPA_crawling.xml")
mj_data = mujoco.MjData(mj_model)

# 2. Plan gait
anchors_a, anchors_b = read_anchors_from_mujoco(mj_model, mj_data)
sched = ContactScheduler(anchors_a=anchors_a, anchors_b=anchors_b)
plan = sched.plan_traversal(start_a=2, start_b=2, n_steps=4)
planner = LocomotionPlanner(sched)
swing = SwingPlanner(sched, clearance=0.08)
```

```
# 3. Initial configuration
q0 = dock_configuration(robot.model,
    sched.anchor_se3('a', 2), sched.anchor_se3('b', 2))
planner.calibrate_from_config(
    robot.update(q0, np.zeros(18)).r_com)

# 4. Build solvers
nmpc = CentroidalNMPC(CentroidalNMPCConfig(
    robot_mass=31.68, N=20, dt=0.05, ...))
nmpc.build()
wbqp = WholeBodyQP(WholeBodyQPConfig(
    nq=12, nc_max=2, dt_qp=0.008, ...))

# 5. Control loop: see simulation_loop.py
```